Solutions

Maths

Q1. Let C be a curve defined as the locus of a point $P(x, y)$ such that its distance from the point $S(2, 1)$ is equal to its perpendicular distance from the line $x = 0$. The area (in sq. units) of the region bounded by the curve C and the line $x = 2$ is:

(A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{4}{3}$ (D) $\frac{10}{3}$

Ans. A

Sol. According to the focus-directrix property ($PS = PM$), the distance of $P(x, y)$ from $S(2, 1)$ equals its distance from the line $x = 0$:

$\sqrt{(x-2)^2 + (y-1)^2} = |x|$

Squaring both sides:

$(x-2)^2 + (y-1)^2 = x^2$

$x^2 - 4x + 4 + (y-1)^2 = x^2$

$(y-1)^2 = 4(x-1)$

This represents a parabola with vertex at $(1, 1)$ opening towards the positive x-axis.

The required area bounded by the parabola and the line $x = 2$ is:

$\text{Area} = \int_1^2 2\sqrt{4(x-1)}\,dx = 4\int_1^2 \sqrt{x-1}\,dx$

$\text{Area} = 4\left[\frac{2}{3}(x-1)^{3/2}\right]_1^2 = \frac{8}{3}$ sq. units

Q2. For $x \in (0, 1)$, let $f(x) = \tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right)$ and $g(x) = \cos^{-1}(2x^2 - 1)$. Then the derivative of $f(x)$ with respect to $g(x)$ at $x = \frac{1}{2}$ is equal to:

(A) $\frac{1}{2\sqrt{5}}$ (B) $\frac{1}{\sqrt{5}}$ (C) $\frac{1}{4\sqrt{5}}$ (D) $\frac{\sqrt{5}}{4}$

Ans. A

Sol. To simplify $f(x)$, substitute $x^2 = \cos 2\theta$, where $\theta \in \left(0, \frac{\pi}{4}\right)$ since $x \in (0, 1)$.

$f(x) = \tan^{-1}\left(\frac{\sqrt{2}\cos\theta + \sqrt{2}\sin\theta}{\sqrt{2}\cos\theta - \sqrt{2}\sin\theta}\right) = \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \theta\right)\right)$

Since $\theta \in \left(0, \frac{\pi}{4}\right)$, we have $\left(\frac{\pi}{4} + \theta\right) \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$.

Thus: $f(x) = \frac{\pi}{4} + \theta = \frac{\pi}{4} + \frac{1}{2}\cos^{-1}(x^2)$

$f'(x) = \frac{1}{2} \cdot \left(-\frac{1}{\sqrt{1-x^4}}\right) \cdot (2x) = -\frac{x}{\sqrt{1-x^4}}$

For $g(x) = \cos^{-1}(2x^2 - 1)$, substitute $x = \cos\phi$, where $\phi \in \left(0, \frac{\pi}{2}\right)$:

$g(x) = \cos^{-1}(2\cos^2\phi - 1) = \cos^{-1}(\cos 2\phi) = 2\phi = 2\cos^{-1} x$

$g'(x) = -\frac{2}{\sqrt{1-x^2}}$

Now, the derivative of $f(x)$ with respect to $g(x)$ is: $\frac{df}{dg} = \frac{f'(x)}{g'(x)} = \frac{-\frac{x}{\sqrt{1-x^4}}}{-\frac{2}{\sqrt{1-x^2}}}$

$= \frac{x}{2}\sqrt{\frac{1-x^2}{1-x^4}} = \frac{x}{2\sqrt{1+x^2}}$

$\left.\frac{df}{dg}\right|_{x=1/2} = \frac{1/2}{2\sqrt{1+\frac{1}{4}}} = \frac{1/2}{2 \cdot \frac{\sqrt{5}}{2}} = \frac{1}{2\sqrt{5}}$

Q3. If a normal to the parabola $y^2 = 8x$ is parallel to the line $x - y + 3 = 0$ and intersects the parabola again at the point Q, then the coordinates of Q are:

(A) $(18, 12)$ (B) $(18, -12)$ (C) $(8, 8)$ (D) $(8, -8)$

Ans. A

Sol. For the parabola $y^2 = 8x$, we have $4a = 8 \implies a = 2$.

The parametric coordinates of any point on the parabola are $(at^2, 2at) = (2t^2, 4t)$.

The slope of the normal to the parabola at parameter t_1 is given by $m = -t_1$.

The line parallel to the normal is $x - y + 3 = 0$, which has slope $m = 1$.

Thus, $-t_1 = 1 \implies t_1 = -1$.

If a normal drawn at parameter t_1 intersects the parabola again at parameter t_2, then:
$t_2 = -t_1 - \frac{2}{t_1}$

$t_2 = -(-1) - \frac{2}{-1} = 1 + 2 = 3$

The coordinates of the point Q corresponding to parameter $t_2 = 3$ are:

$Q = (at_2^2, 2at_2) = (2(3)^2, 2(2)(3)) = (18, 12)$.

Q4. If the curve representing the orthogonal trajectory of the family of curves $y = c(x+1)e^x$ (where c is an arbitrary constant and $x > -1$) passes through the point $(0, 2)$, then its equation is:

(A) $y^2 + 2x - 4 = \ln\left(\dfrac{x+2}{2}\right)^2$ (B) $y^2 - 2x - 4 = \ln\left(\dfrac{x+2}{2}\right)^2$

(C) $y^2 + 2x - 4 = 2\ln(x+2)$ (D) $y^2 - 2x + 4 = \ln\left(\dfrac{x+2}{2}\right)^2$

Ans. A

Sol. The given family of curves is: $y = c(x+1)e^x$

$$\frac{dy}{dx} = ce^x(x+1) + ce^x(1) = ce^x(x+2)$$

$$\frac{dy}{dx} = \frac{y(x+2)}{x+1}$$

To find the differential equation of the orthogonal trajectory, replace $\dfrac{dy}{dx}$ with $-\dfrac{dx}{dy}$:

$$-\frac{dx}{dy} = \frac{y(x+2)}{x+1} \Rightarrow \frac{dy}{dx} = -\frac{x+1}{(x+2)y}$$

$$y\,dy = -\frac{x+1}{x+2}\,dx = -\left(1 - \frac{1}{x+2}\right)dx$$

$$\int y\,dy = \int \left(\frac{1}{x+2} - 1\right)dx$$

$$\frac{y^2}{2} = \ln(x+2) - x + C_1$$

$$y^2 + 2x - 2\ln(x+2) = C$$

Since the curve passes through the point $(0, 2)$: $2^2 + 2(0) - 2\ln(2) = C \implies C = 4 - 2\ln 2$

$$y^2 + 2x - 2\ln(x+2) = 4 - 2\ln 2$$

$$y^2 + 2x - 4 = 2\ln(x+2) - 2\ln 2 = 2\ln\left(\frac{x+2}{2}\right) = \ln\left(\frac{x+2}{2}\right)^2$$

Q5. Consider the differential equation

$$2(x\,dx + y\,dy) = (x^2 + y^2 - 1)(x\,dy - y\,dx)$$

Using the polar substitution $x = r\cos\theta$ and $y = r\sin\theta$, the general solution of the differential equation is given by (where C is an arbitrary constant):

(A) $\ln\left|1 - \dfrac{1}{x^2+y^2}\right| = \tan^{-1}\left(\dfrac{y}{x}\right) + C$ (B) $\ln\left|1 + \dfrac{1}{x^2+y^2}\right| = \tan^{-1}\left(\dfrac{y}{x}\right) + C$

(C) $\ln\left|\dfrac{x^2+y^2}{x^2+y^2-1}\right| = \tan^{-1}\left(\dfrac{y}{x}\right)+C$ (D) $\ln\left|x^2+y^2-1\right| = 2\tan^{-1}\left(\dfrac{y}{x}\right)+C$

Ans. A

Sol. Substitute $x = r\cos\theta$ and $y = r\sin\theta$.

We have $x^2+y^2 = r^2$.

Differentiating gives: $x\,dx + y\,dy = r\,dr$ & $x\,dy - y\,dx = r^2\,d\theta$

Substituting these into the given differential equation:

$2(r\,dr) = (r^2-1)(r^2\,d\theta)$

$\Rightarrow \dfrac{2}{r(r^2-1)}dr = d\theta$

$\displaystyle\int\left(-\frac{2}{r}+\frac{1}{r-1}+\frac{1}{r+1}\right)dr = \int d\theta$

$-2\ln|r| + \ln|r-1| + \ln|r+1| = \theta + C$

$ln\left|\frac{(r-1)(r+1)}{r^2}\right| = \theta + C$

$ln\left|1-\frac{1}{r^2}\right| = \theta + C$

Substituting $r^2 = x^2+y^2$ and $\theta = \tan^{-1}\left(\dfrac{y}{x}\right)$:

$\ln\left|1-\dfrac{1}{x^2+y^2}\right| = \tan^{-1}\left(\dfrac{y}{x}\right)+C$

Q6. The area (in sq. units) of the region bounded by the curve $y = |x^2-4x+3|$ and the line $y = x-1$ is:

(A) $\dfrac{11}{6}$ (B) $\dfrac{13}{6}$ (C) $\dfrac{5}{2}$ (D) $\dfrac{7}{3}$

Ans. B

Sol. First, find the points of intersection between $y = |x^2-4x+3|$ and $y = x-1$.

Case 1: $x \geq 3$ or $x \leq 1$
Here, $x^2-4x+3 = x-1 \implies x^2-5x+4 = 0 \implies x = 1, 4.$

Case 2: $1 < x < 3$

Here, $-(x^2-4x+3) = x-1 \implies x^2-3x+2 = 0 \implies x = 1, 2.$

Thus, the bounded region spans from $x = 1$ to $x = 4$.

We split the integral into three sub-intervals based on relative positions of the curves:

1. For $x \in [1,2]$, $y = -x^2+4x-3 \geq y = x-1$:

$A_1 = \displaystyle\int_1^2 \big((-x^2+4x-3)-(x-1)\big)dx = \int_1^2(-x^2+3x-2)dx = \left[-\frac{x^3}{3}+\frac{3x^2}{2}-2x\right]_1^2 = \frac{1}{6}$

2. For $x \in [2,3]$, $y = x-1 \geq y = -x^2+4x-3$:

$$A_2 = \int_2^3 \big((x-1)-(-x^2+4x-3)\big)dx = \int_2^3 (x^2-3x+2)dx = \left[\frac{x^3}{3}-\frac{3x^2}{2}+2x\right]_2^3 = \frac{5}{6}$$

3. For $x \in [3,4]$, $y = x-1 \geq y = x^2-4x+3$:

$$A_3 = \int_3^4 \big((x-1)-(x^2-4x+3)\big)dx = \int_3^4 (-x^2+5x-4)dx = \left[-\frac{x^3}{3}+\frac{5x^2}{2}-4x\right]_3^4 = \frac{7}{6}$$

Total required area: $A = A_1 + A_2 + A_3 = \frac{1}{6}+\frac{5}{6}+\frac{7}{6} = \frac{13}{6}$

Q7. If the parabola $x^2 = 4ay$ (where $a > 0$) touches the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at two distinct points, then the eccentricity of the hyperbola is equal to:

(A) $\frac{\sqrt{5}}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{2}$ (D) $\frac{\sqrt{6}}{2}$

Ans. A

Sol. From the equation of the parabola, we have $y = \frac{x^2}{4a}$.

Substituting y into the equation of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$:

$$\frac{x^2}{a^2}-\frac{x^4}{16a^2b^2}=1$$

$$x^4 - 16b^2x^2 + 16a^2b^2 = 0$$

Since the parabola and hyperbola touch each other, this quadratic equation in x^2 must have equal roots.

Therefore, the discriminant D must be zero: $D = (-16b^2)^2 - 4(1)(16a^2b^2) = 0$

$$256b^4 - 64a^2b^2 = 0$$

Since $b \neq 0$: $\frac{b^2}{a^2} = \frac{64}{256} = \frac{1}{4}$

The eccentricity e of the hyperbola is given by: $e = \sqrt{1+\frac{b^2}{a^2}} = \sqrt{1+\frac{1}{4}} = \frac{\sqrt{5}}{2}$

Q8. Let $H: \frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ $(a>0, b>0)$ be a hyperbola with eccentricity e. Let S and S' be the foci of H. A circle C_1 is drawn taking SS' as a diameter. Let P be the point of intersection of H and C_1 lying in the first quadrant. If the normal to H at P passes through the point $(0, 2b)$, then the eccentricity e of the hyperbola is equal to:

(A) $\frac{2}{\sqrt{3}}$ (B) $\sqrt{2}$ (C) $\frac{\sqrt{5}}{2}$ (D) $\frac{3}{2}$

Ans. A

Sol. The foci of the hyperbola H are $S(ae, 0)$ and $S'(-ae, 0)$.

Since SS' is the diameter of circle C_1, the center of C_1 is the origin $(0, 0)$ and its radius is $ae = \sqrt{a^2 + b^2}$.

Thus, the equation of circle C_1 is: $x^2 + y^2 = a^2 + b^2$ Let $P(x_1, y_1)$ be the point of intersection in the first quadrant.

Since P lies on both C_1 and H: $x_1^2 + y_1^2 = a^2 + b^2$ --- (1)

$\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$ --- (2)

The equation of the normal to H at $P(x_1, y_1)$ is given by: $\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$

Since this normal passes through $(0, 2b)$: $\frac{b^2(2b)}{y_1} = a^2 + b^2 \implies y_1 = \frac{2b^3}{a^2 + b^2}$

From equation (1), $x_1^2 = (a^2 + b^2) - y_1^2 = (a^2 + b^2) - \frac{4b^6}{(a^2 + b^2)^2}$.

Substituting x_1^2 and y_1 into equation (2): $\frac{a^2 + b^2}{a^2} - \frac{4b^6}{a^2(a^2 + b^2)^2} - \frac{4b^4}{(a^2 + b^2)^2} = 1$

Let $k = \frac{b^2}{a^2}$.

Dividing numerator and denominator by a^2 gives: $(1 + k) - \frac{4k^3}{(1 + k)^2} - \frac{4k^2}{(1 + k)^2} = 1$

$k = \frac{4k^2(k + 1)}{(1 + k)^2} = \frac{4k^2}{1 + k}$

Since $k > 0$: $1 + k = 4k \implies 3k = 1 \implies k = \frac{1}{3}$

Therefore, the eccentricity e is: $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + k} = \sqrt{1 + \frac{1}{3}} = \frac{2}{\sqrt{3}}$

Q9. The rate of change of a quantity y with respect to x is given by $\frac{dy}{dx} = \frac{1}{x(x^3 + 1)}$ for $x > 0$. If $y(1) = 0$, then the value of $y(2)$ is equal to:

(A) $\frac{1}{3}\ln\left(\frac{16}{9}\right)$ (B) $\frac{1}{3}\ln\left(\frac{8}{9}\right)$ (C) $\frac{1}{3}\ln\left(\frac{4}{9}\right)$ (D) $\frac{1}{6}\ln\left(\frac{16}{9}\right)$

Ans. A

Sol. Given the rate measure equation: $dy = \frac{dx}{x(x^3 + 1)}$

$dy = \frac{x^2 dx}{x^3(x^3 + 1)}$

$y = \int \frac{x^2 dx}{x^3(x^3 + 1)}$

Let $t = x^3$, then $dt = 3x^2 dx \implies x^2 dx = \frac{dt}{3}$.

$y = \frac{1}{3}\int \frac{dt}{t(t + 1)} = \frac{1}{3}\int \left(\frac{1}{t} - \frac{1}{t + 1}\right) dt$

$$y = \frac{1}{3}\ln\left|\frac{t}{t+1}\right| + C = \frac{1}{3}\ln\left(\frac{x^3}{x^3+1}\right) + C$$

Given $y(1) = 0$: $0 = \frac{1}{3}\ln\left(\frac{1}{2}\right) + C \implies C = -\frac{1}{3}\ln\left(\frac{1}{2}\right) = \frac{1}{3}\ln(2)$

Thus, $y(x) = \frac{1}{3}\ln\left(\frac{2x^3}{x^3+1}\right)$

Evaluating at $x = 2$: $y(2) = \frac{1}{3}\ln\left(\frac{2(2)^3}{2^3+1}\right) = \frac{1}{3}\ln\left(\frac{16}{9}\right)$

Q10. Let $y(x)$ be the solution of the differential equation $(1+x^2)x\frac{dy}{dx} + (1+2x^2)y = x(1+x^2)^{3/2}$ for $x > 0$, satisfying $y(1) = \frac{3}{4\sqrt{2}}$. Then the value of $y(\sqrt{3})$ is equal to:

(A) $\frac{5\sqrt{3}}{8}$ (B) $\frac{3\sqrt{3}}{8}$ (C) $\frac{5\sqrt{3}}{4}$ (D) $\frac{\sqrt{3}}{2}$

Ans. A

Sol. $x\frac{dy}{dx} + \left(\frac{1+2x^2}{1+x^2}\right)y = x\sqrt{1+x^2}$

$\left(x\frac{dy}{dx} + y\right) + \frac{x^2y}{1+x^2} = x\sqrt{1+x^2}$

$\frac{d(xy)}{dx} + \frac{x}{1+x^2}(xy) = x\sqrt{1+x^2}$

Let $u = xy$.

$\frac{du}{dx} + \frac{x}{1+x^2}u = x\sqrt{1+x^2}$

The Integrating Factor (I.F.) is: I.F. $= e^{\int \frac{x}{1+x^2}dx} = e^{\frac{1}{2}\ln(1+x^2)} = \sqrt{1+x^2}$

$u\sqrt{1+x^2} = \int x\sqrt{1+x^2}\cdot\sqrt{1+x^2}\,dx + C$

$xy\sqrt{1+x^2} = \int (x+x^3)\,dx + C = \frac{x^2}{2} + \frac{x^4}{4} + C$

Using the initial condition $y(1) = \frac{3}{4\sqrt{2}}$: $1\cdot\frac{3}{4\sqrt{2}}\cdot\sqrt{2} = \frac{1}{2} + \frac{1}{4} + C$

$\implies \frac{3}{4} = \frac{3}{4} + C \implies C = 0$

Thus, for $x > 0$: $y\sqrt{1+x^2} = \frac{x(2+x^2)}{4}$

Substitute $x = \sqrt{3}$: $y(\sqrt{3})\sqrt{1+3} = \frac{\sqrt{3}(2+3)}{4} \implies 2y(\sqrt{3}) = \frac{5\sqrt{3}}{4} \implies y(\sqrt{3}) = \frac{5\sqrt{3}}{8}$

Q11. Consider an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a > b > 0)$ whose eccentricity is $\frac{1}{\sqrt{2}}$ and the length of its latus rectum is 4. Let $P(\alpha, \beta)$ be any point lying on the line $x + y = 6$. If the polar of the point P with respect to the ellipse always passes through a fixed point $Q(x_0, y_0)$, then the distance of Q from the origin is:

(A) $\frac{4\sqrt{5}}{3}$ (B) $\frac{2\sqrt{5}}{3}$ (C) $\frac{8\sqrt{5}}{3}$ (D) $\frac{5\sqrt{3}}{4}$

Ans. A

Sol. Given the eccentricity $e = \frac{1}{\sqrt{2}}$, we have: $e^2 = 1 - \frac{b^2}{a^2} \implies \frac{1}{2} = 1 - \frac{b^2}{a^2} \implies a^2 = 2b^2$

The length of the latus rectum is given as 4: $\frac{2b^2}{a} = 4 \implies \frac{a^2}{a} = 4 \implies a = 4$

Thus, $a^2 = 16$ and $b^2 = 8$. The equation of the ellipse is: $\frac{x^2}{16} + \frac{y^2}{8} = 1$

Since $P(\alpha, \beta)$ lies on $x + y = 6$, we have $\alpha + \beta = 6 \implies \beta = 6 - \alpha$.

The equation of the polar of point $P(\alpha, \beta)$ with respect to the ellipse is $T = 0$: $\frac{\alpha x}{16} + \frac{\beta y}{8} = 1$

Substituting $\beta = 6 - \alpha$: $\frac{\alpha x}{16} + \frac{(6-\alpha)y}{8} = 1$

$\alpha\left(\frac{x}{16} - \frac{y}{8}\right) + \left(\frac{3y}{4} - 1\right) = 0$

This represents a family of lines passing through the intersection of: $\frac{3y}{4} - 1 = 0 \implies y = \frac{4}{3}$
& $\frac{x}{16} - \frac{y}{8} = 0 \implies x = 2y = \frac{8}{3}$

Hence, the fixed point is $Q\left(\frac{8}{3}, \frac{4}{3}\right)$.

The distance of Q from the origin is: $OQ = \sqrt{\left(\frac{8}{3}\right)^2 + \left(\frac{4}{3}\right)^2} = \sqrt{\frac{64+16}{9}} = \frac{\sqrt{80}}{3} = \frac{4\sqrt{5}}{3}$

Q12. Let $f : [0, 2] \to \mathbb{R}$ be a function which is continuous on $[0, 2]$ and twice differentiable on $(0, 2)$. If $f(0) = 0$, $f(1) = 3$, and $f(2) = 3$, then which of the following statements MUST be **TRUE** ?

(A) There exists at least one $c \in (0, 2)$ such that $f''(c) < -\frac{3}{2}$

(B) There exists at least one $c \in (0, 2)$ such that $f''(c) = 0$

(C) There exists at least one $c \in (0, 2)$ such that $f''(c) > \frac{3}{2}$ (D) $f'(c) > 0$ for all $c \in (0, 1)$

Ans. A

Sol. Applying Lagrange's Mean Value Theorem (LMVT) to $f(x)$ on the interval $[0, 1]$:

Since $f(x)$ is continuous on $[0, 1]$ and differentiable on $(0, 1)$, there exists some $c_1 \in (0, 1)$ such that $f'(c_1) = \frac{f(1) - f(0)}{1 - 0} = \frac{3 - 0}{1} = 3$

Applying Rolle's Theorem to $f(x)$ on the interval $[1, 2]$:

Since $f(x)$ is continuous on $[1, 2]$, differentiable on $(1, 2)$, and $f(1) = f(2) = 3$, there exists some $c_2 \in (1, 2)$ such that $f'(c_2) = 0$

Now, applying LMVT to the derivative function $f'(x)$ on the interval $[c_1, c_2] \subset (0, 2)$:

Since $f'(x)$ is continuous on $[c_1, c_2]$ and differentiable on (c_1, c_2), there exists some $c \in (c_1, c_2) \subset (0, 2)$ such that $f''(c) = \dfrac{f'(c_2) - f'(c_1)}{c_2 - c_1} = \dfrac{0 - 3}{c_2 - c_1} = -\dfrac{3}{c_2 - c_1}$

Since $c_1 \in (0, 1)$ and $c_2 \in (1, 2)$, the length of interval satisfies: $0 < c_2 - c_1 < 2$ $\implies \dfrac{1}{c_2 - c_1} > \dfrac{1}{2}$

$f''(c) = -\dfrac{3}{c_2 - c_1} < -\dfrac{3}{2}$

Thus, there must exist at least one $c \in (0, 2)$ such that $f''(c) < -\dfrac{3}{2}$.

Q13. If the chord of contact of a point P with respect to the parabola $y^2 = 4x$ is given by $2x - y + 1 = 0$, and the equation of the pair of tangents drawn from P to the ellipse $2x^2 + y^2 = 2$ is written as $ax^2 + bxy + cy^2 + dx + ey + f = 0$ (where a, b, c, d, e, f are integers having no common factor greater than 1 and $a > 0$), then the value of $a + b + c + d + e + f$ is equal to:

(A) 1 (B) 2 (C) 3 (D) 4

Ans. A

Sol. Let the coordinates of point P be (h, k).

The equation of the chord of contact of $P(h, k)$ with respect to the parabola $y^2 = 4x$ is given by $T = 0$: $yk = 2(x + h) \implies 2x - ky + 2h = 0$

Comparing this with the given equation $2x - y + 1 = 0$, we get: $k = 1$ and $2h = 1 \implies h = \dfrac{1}{2}$

Thus, the coordinates of point P are $\left(\dfrac{1}{2}, 1\right)$.

Now, for the ellipse $2x^2 + y^2 = 2$, let $S \equiv 2x^2 + y^2 - 2 = 0$.
The equation of the pair of tangents from P to the ellipse is given by $SS_1 = T^2$:
$(2x^2 + y^2 - 2)\left(-\dfrac{1}{2}\right) = (x + y - 2)^2$

$4x^2 + 4xy + 3y^2 - 8x - 8y + 6 = 0$
$a = 4,\ b = 4,\ c = 3,\ d = -8,\ e = -8,\ f = 6$
Therefore, the required sum is: $a + b + c + d + e + f = 4 + 4 + 3 - 8 - 8 + 6 = 1$

Q14. Let $y = y(x)$ be the solution of the differential equation $(x + y\sqrt{x^2 + y^2})dx + (y - x\sqrt{x^2 + y^2})dy = 0$ satisfying $y(1) = 0$. If $y(x_0) = x_0$ for some $x_0 > 0$, then the value of $x_0^2 + (y(x_0))^2$ is equal to

(A) $\dfrac{16}{(4-\pi)^2}$ (B) $\dfrac{4}{4-\pi}$ (C) $\dfrac{16}{(4+\pi)^2}$ (D) $\dfrac{4}{4+\pi}$

Ans. A

Sol. The given differential equation is: $(x\,dx + y\,dy) + \sqrt{x^2+y^2}(y\,dx - x\,dy) = 0$

Using polar substitution $x = r\cos\theta$ and $y = r\sin\theta$, we have

$r\,dr + r(-r^2\,d\theta) = 0$

$r\,dr - r^3\,d\theta = 0$

$\dfrac{dr}{r^2} = d\theta$

Integrating both sides: $-\dfrac{1}{r} = \theta + C$

Given $y(1) = 0$, which corresponds to $x = 1, y = 0 \implies r = 1, \theta = 0$:

$-\dfrac{1}{1} = 0 + C \implies C = -1$

Thus, the relation becomes: $\dfrac{1}{r} = 1 - \theta \implies r = \dfrac{1}{1-\theta}$

For $y(x_0) = x_0$ with $x_0 > 0$: $\tan\theta = \dfrac{y(x_0)}{x_0} = 1 \implies \theta = \dfrac{\pi}{4}$

So, the value of r at $x = x_0$ is: $r = \dfrac{1}{1-\frac{\pi}{4}} = \dfrac{4}{4-\pi}$

We are required to find $x_0^2 + (y(x_0))^2 = r^2$: $r^2 = \left(\dfrac{4}{4-\pi}\right)^2 = \dfrac{16}{(4-\pi)^2}$

Q15. Let $f(x) = \displaystyle\int_0^x (t - \lfloor t \rfloor)\,dt$ for $x > 0$, where $\lfloor\cdot\rfloor$ denotes the greatest integer function. If $g(x) = \displaystyle\int_0^{f(x)} \frac{1}{1+t^2}\,dt$, then the value of $g'\left(\dfrac{3}{2}\right)$ is equal to:

(A) $\dfrac{32}{89}$ (B) $\dfrac{16}{89}$ (C) $\dfrac{32}{65}$ (D) $\dfrac{16}{65}$

Ans. A

Sol. First, we evaluate $f\left(\dfrac{3}{2}\right)$ by breaking the integral at points of discontinuity:

$f\left(\dfrac{3}{2}\right) = \displaystyle\int_0^1 t\,dt + \int_1^{3/2} (t-1)\,dt$

$= \left[\dfrac{t^2}{2}\right]_0^1 + \left[\dfrac{(t-1)^2}{2}\right]_1^{3/2} = \dfrac{1}{2} + \dfrac{1}{8} = \dfrac{5}{8}$

Since $x = \dfrac{3}{2}$ is not an integer, the integrand $(t - \lfloor t \rfloor)$ is continuous at $t = \dfrac{3}{2}$.

Applying the Leibniz rule to $f(x)$: $f'\left(\dfrac{3}{2}\right) = \dfrac{3}{2} - \left\lfloor\dfrac{3}{2}\right\rfloor = \dfrac{3}{2} - 1 = \dfrac{1}{2}$

Now, using the Leibniz rule to differentiate $g(x)$:

$g'(x) = \dfrac{d}{dx}\displaystyle\int_0^{f(x)} \frac{1}{1+t^2}\,dt = \frac{1}{1+(f(x))^2}\cdot f'(x)$

Substituting $x = \frac{3}{2}$: $g'\left(\frac{3}{2}\right) = \frac{1}{1+\left(f\left(\frac{3}{2}\right)\right)^2} \cdot f'\left(\frac{3}{2}\right)$

$= \frac{1}{1+\left(\frac{5}{8}\right)^2} \cdot \frac{1}{2} = \frac{1}{1+\frac{25}{64}} \cdot \frac{1}{2}$

$= \frac{64}{89} \cdot \frac{1}{2} = \frac{32}{89}$

Q16. Let $y = f(x)$ be the solution of the differential equation

$$\frac{xdy - ydx}{\sqrt{x^2 - y^2}} = dx \quad (x > 0)$$

satisfying the initial condition $f(1) = 0$. If m and n are the order and degree, respectively, of the differential equation corresponding to the family of curves $y = C_1 f(x) + C_2 f'(x)$ where C_1 and C_2 are arbitrary constants, then the value of $m + 2n$ is equal to:

(A) 3 (B) 4 (C) 5 (D) 6

Ans. B

Sol. Given differential equation: $\frac{xdy - ydx}{\sqrt{x^2 - y^2}} = dx$

$$\frac{\frac{xdy-ydx}{x^2}}{\frac{1}{x}\sqrt{x^2 - y^2}} = dx \implies \frac{d\left(\frac{y}{x}\right)}{\sqrt{1-\left(\frac{y}{x}\right)^2}} = \frac{dx}{x}$$

Integrating both sides:

$\arcsin\left(\frac{y}{x}\right) = \ln x + C$

Given $f(1) = 0 \implies y = 0$ when $x = 1$: $\arcsin(0) = \ln(1) + C \implies C = 0$

Thus, $\frac{y}{x} = \sin(\ln x) \implies f(x) = x\sin(\ln x)$

Differentiating $f(x)$ with respect to x:

$f'(x) = \sin(\ln x) + x \cdot \frac{1}{x}\cos(\ln x) = \sin(\ln x) + \cos(\ln x)$

Now, the given family of curves is: $y = C_1 x\sin(\ln x) + C_2(\sin(\ln x) + \cos(\ln x))$

Since there are 2 independent arbitrary constants (C_1 and C_2), the order of the corresponding differential equation is $m = 2$.

Since C_1 and C_2 occur linearly in the family of curves, eliminating them yields a linear differential equation in y and its derivatives, which means the degree of the highest derivative is $n = 1$.

Therefore, $m + 2n = 2 + 2(1) = 4$

Q17. Let $f : [0, 4] \to \mathbb{R}$ be a continuous function on $[0, 4]$ and differentiable on $(0, 4)$ such that $f(0) = 1$, $f(1) = 3$, $f(2) = 0$, $f(3) = 4$, and $f(4) = 1$. The minimum number of roots of the equation $f'(x) = 0$ in the interval $(0, 4)$ is:

(A) 2 (B) 3 (C) 4 (D) 1

Ans. B

Sol. Define an auxiliary function $g(x) = f(x) - 1$.

Evaluating $g(x)$ at the given points:
$g(0) = f(0) - 1 = 0$; $g(1) = f(1) - 1 = 2 > 0$; $g(2) = f(2) - 1 = -1 < 0$;
$g(3) = f(3) - 1 = 3 > 0$; $g(4) = f(4) - 1 = 0$

By the Intermediate Value Theorem (IVT) applied to $g(x)$:
1. Since $g(1) > 0$ and $g(2) < 0$, there exists at least one $\alpha \in (1, 2)$ such that $g(\alpha) = 0$.
2. Since $g(2) < 0$ and $g(3) > 0$, there exists at least one $\beta \in (2, 3)$ such that $g(\beta) = 0$.

Thus, $g(x) = 0$ at $x = 0, \alpha, \beta, 4$ where $0 < \alpha < 2 < \beta < 4$.

Applying Rolle's Theorem to $g(x)$ on each sub-interval:
On $[0, \alpha]$: $g(0) = g(\alpha) = 0 \implies \exists c_1 \in (0, \alpha)$ such that $g'(c_1) = f'(c_1) = 0$.
On $[\alpha, \beta]$: $g(\alpha) = g(\beta) = 0 \implies \exists c_2 \in (\alpha, \beta)$ such that $g'(c_2) = f'(c_2) = 0$.
On $[\beta, 4]$: $g(\beta) = g(4) = 0 \implies \exists c_3 \in (\beta, 4)$ such that $g'(c_3) = f'(c_3) = 0$.

Since $0 < c_1 < \alpha < c_2 < \beta < c_3 < 4$, the points c_1, c_2, c_3 are distinct and lie in $(0, 4)$.
Hence, $f'(x) = 0$ has at least 3 roots in $(0, 4)$.

Q18. If $\{x\}$ denotes the fractional part of x, then the value of the definite integral $\displaystyle\int_0^{60} \frac{3^{\{x\}}}{3^{\{x\}} + 3^{\{-x\}}}\, dx$ is equal to:

(A) 15 (B) 30 (C) 60 (D) 0

Ans. B

Sol. Let $f(x) = \dfrac{3^{\{x\}}}{3^{\{x\}} + 3^{\{-x\}}}$.
Since $\{x + 1\} = \{x\}$ and $\{-x - 1\} = \{-x\}$, the function $f(x)$ is periodic with period $T = 1$.
Using the property of periodic functions $\displaystyle\int_0^{nT} f(x)\, dx = n \int_0^T f(x)\, dx$, we get:
$I = \displaystyle\int_0^{60} \frac{3^{\{x\}}}{3^{\{x\}} + 3^{\{-x\}}}\, dx = 60 \int_0^1 \frac{3^{\{x\}}}{3^{\{x\}} + 3^{\{-x\}}}\, dx$
For $x \in (0, 1)$, we have $\{x\} = x$ and $\{-x\} = 1 - x$.
Thus,
$I = 60 \displaystyle\int_0^1 \frac{3^x}{3^x + 3^{1-x}}\, dx$
Let $J = \displaystyle\int_0^1 \frac{3^x}{3^x + 3^{1-x}}\, dx$.
Applying the property $\displaystyle\int_0^a g(x)\, dx = \int_0^a g(a - x)\, dx$:

$$J = \int_0^1 \frac{3^{1-x}}{3^{1-x}+3^x}\,dx$$

Adding the two equations for J:

$$2J = \int_0^1 \frac{3^x+3^{1-x}}{3^x+3^{1-x}}\,dx = \int_0^1 1\,dx = 1 \implies J = \frac{1}{2}$$

Hence, $I = 60 \times J = 60 \times \frac{1}{2} = 30$

Q19. Let $E_1 : \frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$ $(a > b > 0)$ be an ellipse with eccentricity e_1. From any point P on the director circle of E_1, tangents PA and PB are drawn to E_1, where A and B are the points of contact. As P moves along the director circle, the chord of contact AB always touches a fixed ellipse $E_2 : \frac{x^2}{\alpha^2}+\frac{y^2}{\beta^2} = 1$ $(\alpha > \beta > 0)$ having eccentricity e_2. If $e_1 = \frac{1}{2}$, then the eccentricity e_2 of E_2 is equal to:

(A) $\frac{\sqrt{5}}{4}$ (B) $\frac{\sqrt{7}}{4}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{7}}{2\sqrt{2}}$

Ans. B

Sol. The equation of the director circle of E_1 is $x^2+y^2 = a^2+b^2$.
Let $P(x_0, y_0)$ be any point on the director circle, so $x_0^2+y_0^2 = a^2+b^2$.

The equation of the chord of contact AB with respect to E_1 is: $\frac{xx_0}{a^2}+\frac{yy_0}{b^2} = 1$

The condition for a line $Ax+By = 1$ to be tangent to the ellipse $E_2 : \frac{x^2}{\alpha^2}+\frac{y^2}{\beta^2} = 1$ is $A^2\alpha^2+B^2\beta^2 = 1$.

Here $A = \frac{x_0}{a^2}$ and $B = \frac{y_0}{b^2}$, so: $\frac{x_0^2\alpha^2}{a^4}+\frac{y_0^2\beta^2}{b^4} = 1$

Since $x_0^2+y_0^2 = a^2+b^2$, we have $\frac{x_0^2}{a^2+b^2}+\frac{y_0^2}{a^2+b^2} = 1$.

Comparing the two identities yields: $\alpha^2 = \frac{a^4}{a^2+b^2}$ and $\beta^2 = \frac{b^4}{a^2+b^2}$

The eccentricity e_2 of E_2 is given by: $e_2^2 = 1-\frac{\beta^2}{\alpha^2} = 1-\frac{b^4}{a^4} = 1-\left(1-e_1^2\right)^2 = 2e_1^2-e_1^4$

Given $e_1 = \frac{1}{2}$: $e_2^2 = 2\left(\frac{1}{4}\right)-\frac{1}{16} = \frac{1}{2}-\frac{1}{16} = \frac{7}{16} \implies e_2 = \frac{\sqrt{7}}{4}$

Q20. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function satisfying $f(x+3) = f(x)$ for all $x \in \mathbb{R}$. Define a function $g(x) = \int_0^{x^2} f(t)\,dt$. If $g'(1) = 6$, then the value of $g'(2)$ is equal to:

(A) 6 (B) 12 (C) 18 (D) 24

Ans. B

Sol. Using Leibniz Rule for differentiation under the integral sign:

$$g'(x) = \frac{d}{dx}\left(\int_0^{x^2} f(t)\,dt\right) = f(x^2)\cdot\frac{d}{dx}(x^2)-f(0)\cdot\frac{d}{dx}(0) = 2xf(x^2)$$

Given $g'(1) = 6$: $g'(1) = 2(1)f(1^2) = 2f(1) = 6 \implies f(1) = 3$

Now, for $x = 2$: $g'(2) = 2(2)f(2^2) = 4f(4)$

Since $f(x)$ is a periodic function with period 3, we have: $f(4) = f(1+3) = f(1) = 3$

Substituting $f(4) = 3$: $g'(2) = 4 \times 3 = 12$

Q1. Let $y = y(x)$ be the solution of the differential equation

$x^2(x\,dx + y\,dy) = (x^2 + y^2)(x\,dy - y\,dx)$

satisfying the condition $y(1) = 0$. If $x_0 > 0$ such that $y(x_0) = x_0 \ln 3$, then the value of $x_0^2 + (y(x_0))^2$ is equal to _____.

Ans. 9

Sol. The given differential equation is: $x^2(x\,dx + y\,dy) = (x^2 + y^2)(x\,dy - y\,dx)$

$$\frac{x\,dx + y\,dy}{x^2 + y^2} = \frac{x\,dy - y\,dx}{x^2}$$

$$\frac{1}{2}d\left(\ln(x^2 + y^2)\right) = d\left(\frac{y}{x}\right)$$

Integrating both sides: $\frac{1}{2}\ln(x^2 + y^2) = \frac{y}{x} + C$

Applying the initial condition $y(1) = 0$: $\frac{1}{2}\ln(1^2 + 0^2) = \frac{0}{1} + C \implies C = 0$

Thus, the implicit solution is: $\ln(x^2 + y^2) = \frac{2y}{x} \implies x^2 + y^2 = e^{\frac{2y}{x}}$

For $x = x_0$, we are given $y(x_0) = x_0 \ln 3$, which implies $\frac{y(x_0)}{x_0} = \ln 3$.

Substituting this into the equation: $x_0^2 + (y(x_0))^2 = e^{2\ln 3} = e^{\ln(3^2)} = 9$

Q2. If A is the area of the region bounded by the curves $y = |x - 1|$ and $y = 3 - |x|$, then the value of A is equal to ____.

Ans. 4

Sol. To find the points of intersection of $y = |x - 1|$ and $y = 3 - |x|$:

For $x \geq 1$: $x - 1 = 3 - x \implies 2x = 4 \implies x = 2$

For $x \leq 0$: $1 - x = 3 + x \implies 2x = -2 \implies x = -1$

For $0 < x < 1$: $1 - x = 3 - x \implies 1 = 3$ (No solution)

Thus, the curves intersect at $x = -1$ and $x = 2$.

In the interval $[-1, 2]$, $3 - |x| \geq |x - 1|$.

The bounded area A is given by: $A = \int_{-1}^{2} ((3 - |x|) - |x - 1|)dx$

Splitting the integral across critical points $x = 0$ and $x = 1$:

$$A = \int_{-1}^{0} ((3+x) - (1-x))dx + \int_{0}^{1} ((3-x) - (1-x))dx + \int_{1}^{2} ((3-x) - (x-1))dx$$

$$A = \int_{-1}^{0} (2+2x)dx + \int_{0}^{1} 2dx + \int_{1}^{2} (4-2x)dx$$

$$A = 1 + 2 + 1 = 4$$

Q3. If A is the area (in sq. units) of the region bounded by the curves $y = |x^2 - 4|$ and $y = 5$, then the value of $3A$ is equal to _____.

Ans. 44

Sol. To find the points of intersection of the curves $y = |x^2 - 4|$ and $y = 5$:

$|x^2 - 4| = 5 \implies x^2 - 4 = 5 \implies x^2 = 9 \implies x = \pm 3$
(Note that $x^2 - 4 = -5 \implies x^2 = -1$, which gives no real solution).

Due to symmetry about the y-axis, the total area A is given by: $A = 2\int_{0}^{3} (5 - |x^2 - 4|)\, dx$

Thus, $3A = 44$.

Q4. A sphere of radius R is expanding such that its radius increases at a constant rate of $2\ \text{cm/s}$. At any instant, a right circular cylinder of maximum volume is inscribed in this sphere. If the rate of change of the maximum volume of the cylinder at the instant when $R = 6\ \text{cm}$ is $k\sqrt{3}\pi\ \text{cm}^3/\text{s}$, then the value of k is equal to ____.

Ans. 96

Sol. Let r and h be the radius and height of the cylinder inscribed in the sphere of radius R.

From geometry, $r^2 + \left(\frac{h}{2}\right)^2 = R^2 \implies r^2 = R^2 - \frac{h^2}{4}$

The volume V of the cylinder is given by:

$$V = \pi r^2 h = \pi \left(R^2 - \frac{h^2}{4}\right)h = \pi \left(R^2 h - \frac{h^3}{4}\right)$$

For maximum volume, $\frac{dV}{dh} = 0$: $\frac{dV}{dh} = \pi \left(R^2 - \frac{3h^2}{4}\right) = 0 \Rightarrow h = \frac{2R}{\sqrt{3}}$

Also, $\frac{d^2V}{dh^2} = -\frac{3\pi h}{2} < 0$, ensuring maximum volume.

Substituting $h = \frac{2R}{\sqrt{3}}$ into V:

$$V_{max} = \pi \left(R^2 \cdot \frac{2R}{\sqrt{3}} - \frac{1}{4} \cdot \frac{8R^3}{3\sqrt{3}}\right) = \frac{4\pi R^3}{3\sqrt{3}}$$

$$\frac{dV_{max}}{dt} = \frac{4\pi}{3\sqrt{3}} \cdot 3R^2 \frac{dR}{dt} = \frac{4\pi R^2}{\sqrt{3}} \frac{dR}{dt}$$

Given $R = 6$ cm and $\frac{dR}{dt} = 2$ cm/s: $\frac{dV_{max}}{dt} = \frac{4\pi(6)^2(2)}{\sqrt{3}} = \frac{288\pi}{\sqrt{3}} = 96\sqrt{3}\pi \text{ cm}^3/\text{s}$

Comparing with $k\sqrt{3}\pi$, we get: $k = 96$

Q5. Let $y = f(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{y-2}{2x+6}$ passing through the point $(0, 5)$. If A is the area of the region bounded by the curve $y = f(x)$ and the line $x = 0$, then find the value of A.

Ans. 12

Sol. To reduce the given differential equation to homogeneous form, substitute $x = X - 3$ and $y = Y + 2$:

$$\frac{dY}{dX} = \frac{Y}{2X}$$

This is a homogeneous differential equation.

$$\int \frac{dY}{Y} = \frac{1}{2}\int \frac{dX}{X} \implies \ln|Y| = \frac{1}{2}\ln|X| + \ln C \implies Y^2 = cX$$

Substituting back $X = x + 3$ and $Y = y - 2$:$(y-2)^2 = c(x+3)$

Since the curve passes through $(0, 5)$:$(5-2)^2 = c(0+3) \implies 9 = 3c \implies c = 3$

Thus, the equation of the curve is:$(y-2)^2 = 3(x+3) \implies x = \frac{(y-2)^2}{3} - 3$

The curve intersects the line $x = 0$ at: $\frac{(y-2)^2}{3} - 3 = 0 \implies (y-2)^2 = 9 \implies y = -1$ and $y = 5$

The area A bounded by the curve and $x = 0$ is: $A = \int_{-1}^{5} (0 - x)\, dy = \int_{-1}^{5} \left(3 - \frac{(y-2)^2}{3}\right) dy$

Let $t = y - 2 \implies dt = dy$: $A = \int_{-3}^{3} \left(3 - \frac{t^2}{3}\right) dt = 2\left[3t - \frac{t^3}{9}\right]_0^3 = 2\,(9 - 3) = 12$